ForSeizure - Software Design Document - Ananth, Parnitha, Salman

Statement of Goals

One in 26 people around the world has epilepsy. For these individuals, predicting when they may have seizures can be a life-changing safety measure. By tracking seizure logs, identifying triggers, and predicting high-risk days, our app will be able to keep people with epilepsy safer.

Functional Description – Minimum Viable Product (MVP)

The application is designed to help users with seizure disorders understand, predict, and manage their condition by tracking potential triggers, forecasting seizure risk, and providing support resources.

Core Features:

1. Trigger Data Collection - this is our MVP (being able to identify triggers)
    - Logs lifestyle, environmental, and physiological factors such as sleep, stress, diet, hormonal changes, and exposure to flashing lights.
    - Uses data analysis to identify patterns and correlations between these factors and seizure occurrences.
2. Seizure Prediction
    - Uses a predictive mathematical model to estimate seizure likelihood based on historical data and current risk factors.
    - Sends timely alerts or notifications to prepare users for high-risk periods.
3. Support Resources
    - Offers guidance, educational content, and links to medical or community resources.
    - May include recommendations for lifestyle adjustments or coping strategies during high-risk periods.

Technical and Data Feasibility

Our application needs to collect both user-reported data and external/biological data:

A. Seizure and Symptom Tracking

- Day, time, and duration of seizures
- Type of seizure and symptoms (e.g., tonic-clonic, absence)
- Aura or pre-seizure signs

B. Lifestyle and Environmental Factors

- Sleep patterns (hours, quality, consistency)
- Stress levels (self-reported or sensor-derived)
- Alcohol and drug consumption (type, amount, frequency)

- Diet (food type, portion, meal timing)
- Menstrual cycle tracking and hormone changes
- Illness or infection records
- Exposure to potential triggers (e.g., flashing lights, environmental stimuli)

C. Medication and Treatment Data

- Type, dosage, and timing of medications

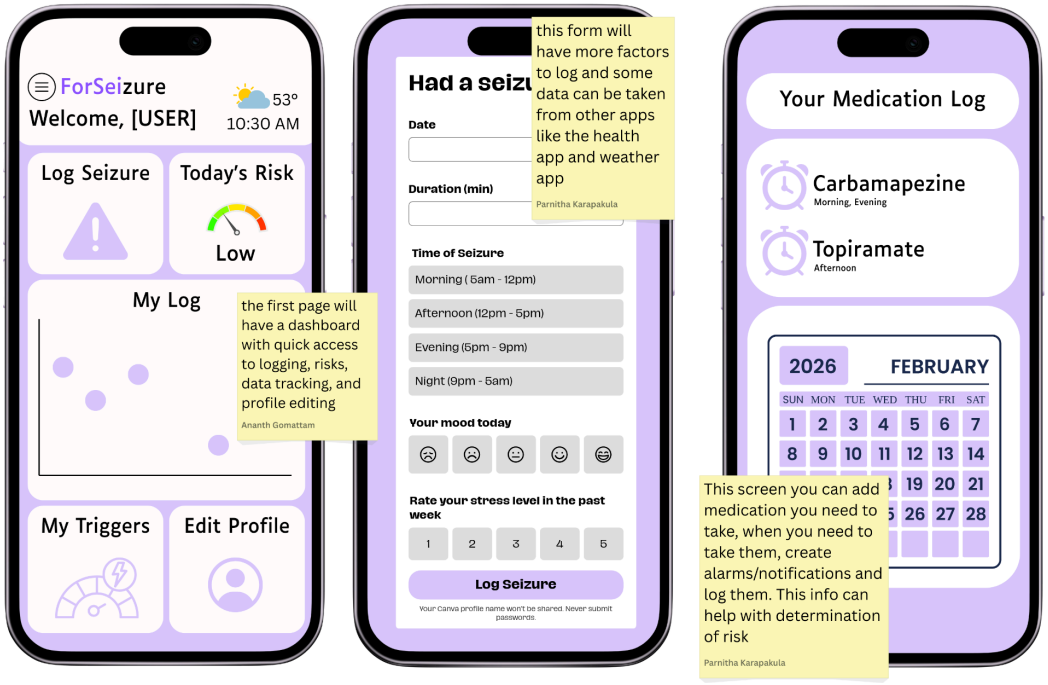
D. Predictive/Biological Data

- Seizure risk predictions (Meta Predict or similar predictive models)

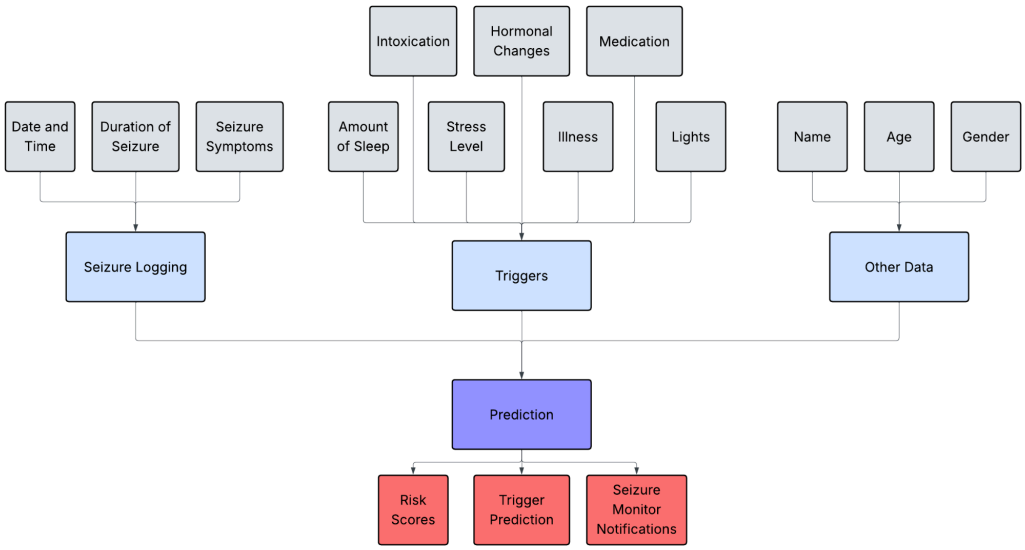
Dependencies / Existing Technologies

- Predictive analytics APIs: Meta Predict for seizure likelihood forecasts
- Database & cloud storage: For securely storing user input and sensor data
- Wearable integrations: Fitbit, Apple Health, or Google Fit for sleep and activity data
- Data visualization libraries: To show trends, correlations, or alerts
- Notification/alert systems: To alert users when triggers are detected or high-risk periods occur

User Interface



Flow Chart and/or Structural Diagram



Persistent Storage

Data stored includes:

- Seizure Logs: data/time, duration, symptoms, etc
- Triggers: sleep, stress, illness, lights, intoxication, hormonal
- Predictions: risk scores, detected triggers
- Basic Settings: notifications and privacy settings

The application will require an input, which is the initial value for each parameter we need for the app to work. After the user starts using the app, it could update depending on what it observes. The application will likely require the use of sensors.

Storage Model:

- Local: recent logs, offline entry, encrypted
- Cloud: long-term history, cross-device access, backups

Privacy & Security:

- Encrypted storage/transfer
- User authentication
- User control over exports, any information can be deleted from the system whenever

Proposed Algorithms:

Mathematical Approach Before Sufficient Data Accumulation

Step 1: Prepare Dataset

- Each day = one cluster
- Each feature = one data point

Step 2: Feature analysis

- For each possible trigger, we can analyze the standard deviation.
- For example,
    - Sleep → 4hrs, 5hrs, 10hrs, 8hrs (seizures) 6hrs, 4,hrs, 8hrs (Normal)
- We will check the standard deviation to see if there is high variance between seizure days→ if yes, it's likely not a trigger
- If we see that the standard deviation is low, we then compare it with 'normal days.'
    - If there isn't a significant difference → likely not a trigger
    - If a significant difference is found → trigger

Step 3: Model refining

- Use other mathematical models or integrate randomness factors
- Employ an ANOVA test on the data instead to analyze interaction effects and causation effects
- Generate p-values to see if the specific "seizure" data point is significantly different

Step 4: Model Improvement

- Gathers a list of triggers and assigns weights to each trigger
- If the trigger is present, the safety score can be calculated by multiplying the trigger's prevalence or scale by the weight
    - Example: Extreme temperatures are identified as a trigger
        - Calculate the deviation from the "normal" value
        - Multiply by weight
        - Add to total score
- The safety score can be calculated by summing all of the trigger risk scores

AI Model Approach After Sufficient Data Accumulation

Large number of logged days, multiple seizure and non-seizure days, consistent tracking of sleep, stress, medication, triggers, etc.

Step 1: Prepare the Dataset

- Each day = one data point
- Inputs → sleep hours, stress level, medication adherence, illness, triggers, past seizure history, rolling averages
- Output label → 1 = seizure occurred, 0 = no seizure

Step 2: Feature Engineering

- Add rolling averages (last 3–7 days sleep, stress)
- Add lag features (seizure yesterday, last week)
- Combine related factors (low sleep + high stress)

Step 3: Train the Advanced Model
- Use Random Forest, Neural Network, maybe an API Key for a pretrained model

Step 4: Model Output
- The model predicts the probability of a seizure for each day
- Example: 0.65 = 65% seizure likelihood
- Safety Score = 1 − probability
- 0.35 safety score = higher risk day

Step 5: Improvement
- Retrain the model periodically with new data
- Personalize per user
- Adjust alert threshold based on user sensitivity

Outside Data Source Identification for Model Training:
If user data is not obtainable within the time frame, outside data sources with seizure causes and triggers are identified. This data can be used to train a general model that can be customized or weighted differently for each user.

- SeizureTracker.com
- NeuroVista
- Empatica
- Human Epilepsy Project
- Neuropace